BY ORDER OF THE COMMANDER
AIR COMBAT COMMAND

AIR COMBAT COMMAND
INSTRUCTION 21-118

28 AUGUST 2026

MAINTENANCE

LOGISTICS MAINTENANCE PERFORMANCE INDICATOR REPORTING PROCEDURES

COMPLIANCE WITH THIS PUBLICATION IS MANDATORY

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This publication implements Department of the Air Force Policy Directive (DAFPD) 21-1, *Maintenance of Military Materiel*. It establishes the requirements and dictates the procedures for reporting aircraft Logistics Performance Indicators (LPI) for assigned aircraft and systems. This instruction applies to all military, civilian, and contractor personnel assigned to Air Combat Command (ACC) units. It also applies to units owning aircraft for which ACC serves as the lead Major Command (MAJCOM), including Classically Associated Air Force Reserve Command (AFRC) and Air National Guard (ANG) units. It does not apply to the United States Space Force. This instruction prescribes the use of the ACC/A4M 9302, *Monthly Maintenance Performance Indicators Report*. Ensure all records created as a result of processes prescribed in this publication are maintained in accordance with (IAW) AFI 33-322, *Records Management and Information Governance Program*, and disposed of IAW the Air Force Records Information Management System Records Disposition Schedule. Contact supporting records managers as required. The authorities to waive wing or unit-level requirements in this publication are identified with a Tier number (e.g., "T-0, T-1, T-2, T-3") following the compliance statement. Refer to Department of the Air Force Manual (DAFMAN) 90-161, *Publishing Processes and Procedures*, for a description of the authorities associated with Tier numbers. Submit waiver requests through the chain of command to the appropriate Tier waiver approval authority, or to the requestor's commander for non-tiered compliance items. Submit comments, questions, and suggested improvements regarding this publication on DAF Form 847, *Recommendation for Change of Product*, through appropriate channels to the HQ ACC/A4MO organizational inbox at **acca4mo@us.af.mil**. This publication may be supplemented at any level, but all supplements must be routed to the OPR of this publication for coordination prior to certification and approval. The use of the name or mark

of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the DAF.

SUMMARY OF CHANGES

This instruction has been revised to incorporate Readiness Informed Metric (RIM) principles, which prioritize maintenance tasks and identify key factors impacting operational performance. It introduces Health of Fleet (HOF) requirements for both the MAJCOM and subordinate units to ensure alignment and consistency across all echelons. Finally, this revision mandates the implementation of control charts, providing a standardized, data-driven framework to monitor performance, identify emerging trends, and drive continuous improvement efforts based on objective analytics.

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Chapter 1

GENERAL INSTRUCTIONS

1.1. Introduction and Purpose. The purpose of this instruction is to establish a standardized, fact-based approach for evaluating logistics performance and guiding decision-making across all ACC units and units reporting to ACC as the lead command. This includes units utilizing Contractor Logistics Support (CLS) and Government CLS partnerships. This instruction provides a consistent framework for documenting the effectiveness of logistics processes that directly impact combat capability, whether executed by organic unit personnel or through contracted services. The primary objectives are to identify areas for improvement in logistics operations, inform resource allocation decisions, and drive continuous improvement initiatives across the command. Recognizing that existing CLS contracts may pose limitations, units will identify any non-compliant portions of their contracts and request waivers IAW DAFMAN 90-161. Command policy prioritizes reducing these waivers through contract renegotiation, and contracting teams will strive to incorporate the intent of this instruction into future agreements. Ultimately, this instruction seeks to optimize logistics operations to maximize combat readiness, pilot upgrade training, and support for real-world contingencies across the command.

1.2. Responsibilities. Wing, Maintenance Group (MXG), Operations Group (OG), and subordinate unit commanders share the responsibility for ensuring compliance with this instruction. MXG Commanders, or their designated representatives, are accountable for the timely and accurate preparation and transmission of all reports outlined herein.

1.2.1. HQ ACC Staff (HQ ACC/A4MO). As the Office of Primary Responsibility (OPR), ACC/A4MO is responsible for providing overarching policy guidance, monitoring command-wide logistics performance indicators, and analyzing HOF data. Additionally, ACC/A4MO evaluates waiver requests and oversees the integration of these reporting procedures across the command to ensure optimized combat readiness.

1.2.2. The Maintenance Management Section, or its equivalent contractor entity, is primarily responsible for developing these reports, regardless of the specific organizational structure. Furthermore, the MXO Superintendent (or equivalent) will maintain direct oversight of this process to ensure the delivery of high-quality, insightful analysis within the comments section of each report. **(T-2)**

1.3. Algorithms. Formulas used to compute rates and performance indicators are listed in Technical Order (TO) 00-20-2, *Maintenance Data Documentation*, or in **Attachment 2** of this instruction**.**

1.4. References. Personnel responsible for generating and analyzing the reports described in this instruction must be familiar with all applicable publications, as well as all applicable command supplements. AFI 10-601, *Operational Capability Requirements Documentation and Validation;* DAFI 21-101, *Aircraft and Equipment Maintenance Management;* DAFI 21-103, *Equipment Inventory, Status, and Utilization Reporting;* AFCSM 21-556 thru 579, *Applicable Air Force Computer Systems Manuals;* DAFMAN 23-300, *Materiel Management Procedures;* DAFPD 10-9, *Lead Command/Lead Agent Designation and Responsibilities for United States Air Force Weapon Systems, Non-Weapon Systems, and Activities.*

Chapter 2

9302 REPORT TRANSMISSION INSTRUCTIONS

2.1. Report Transmission Methods. To support ongoing efforts to modernize and streamline reporting, the HQ ACC/A4MO *Monthly Maintenance Performance Indicators Report* (RCS: ACC-A4MO 9302) template and submission process are subject to periodic updates based on evolving data requirements.

2.1.1. Personnel will obtain the most current guidance and transmission instructions directly from the HQ ACC/A4MO SharePoint Site at: **https://usaf.dps.mil/sites/ACC-A4/A4M/A4MO/SitePages/ACC-A4MO.aspx**. **(T-2)**

2.1.2. In the event the SharePoint site is inaccessible, units will submit reports via email to the organizational inbox at **acca4mo@us.af.mil**. **Note:** For immediate assistance regarding transmission issues, units may contact HQ ACC/A4MO at DSN 574-4810.

2.2. Restrictions on Types of Information.

2.2.1. Units will ensure all submitted reports comply with information security protocols. The standard 9302 report is authorized to contain up to Controlled Unclassified Information (CUI) only. Units will not include Classified information or Personally Identifiable Information (PII) in the standard report submission.

2.2.2. In the rare event that a specific report requires the inclusion of Classified information or PII, ACC/A4MO will specify an appropriate, secure transmission method (typically via SIPRNET email). Units will contact ACC/A4MO at DSN 574-4810 to request the current SIPRNET account address prior to transmission.

Chapter 3

AIRCRAFT REPORTING REQUIREMENTS

3.1. Monthly Logistics Performance Indicators Report. The Monthly Maintenance Performance Indicators Report (RCS: ACC-A4MO 9302) serves as the primary instrument for assessing fleet health across the command.

3.2. Report Use. MAJCOM personnel utilize this report to analyze fleet trends, assess overall fleet health, and determine areas requiring MAJCOM-level assistance or intervention. Furthermore, it serves as an official source document for briefings presented to the ACC Commander and ACC/A4. **(T-2)**

3.3. 9302 Reporting Methods and Instructions. Units will complete the 9302 report via the approved automated application.

3.3.1. HQ ACC/A4MO maintains the official links to the reporting application and current submission instructions on the A4MO SharePoint site: **https://usaf.dps.mil/sites/ACC-A4/A4M/A4MO/SitePages/ACC-A4MO.aspx**. **(T-2)**

3.3.2. **Process Compliance.** Reporting requirements and application interfaces are subject to evolution. Units will refer to the A4MO SharePoint site prior to each reporting cycle to ensure compliance with the most current procedures, system access requirements, and data entry standards. **(T-2)**

3.3.3. **Technical Support and Errors.** If application errors are identified or technical assistance is required, units will contact ACC/A4MO at DSN 574-4810 or via emails at **acca4mo@us.af.mil** **(T-2)**

3.4. Procedures. Units possessing more than one Mission Design Series (MDS) will delineate data for each MDS separately within the report. Units will submit only one comprehensive 9302 report containing all data; separate transmissions for each MDS are not required. Furthermore, units with multiple mission sets (e.g., Operational [OPS], Training [TNG], or Test [TES]) will separate data by mission type. **(T-2)**

3.4.1. **Exceptions.** The following MDS will have missions and blocks combined rather than separated: MQ-1/9, R/T/WC-135, RQ-4, U-2S, and TU-2S. **(T-2)**

3.4.2. F-15C/D series will be reported together; F-15E and F-15EX will be reported separately. F-16 units with multiple blocks will break out the blocks in separate columns. **(T-2)**

3.5. Report Auto-Calculations and Manual Entries. Users will complete the 9302 report utilizing a combination of manual inputs and automatic calculations. The "white" cells require manual data entry, which in turn feeds the auto-calculations in the "grey" cells. Additionally, manual inputs are required at the bottom of Section 1 to identify the name, rank, and telephone number of the unit Point of Contact (POC) responsible for report preparation.

3.6. Home Base Units. Home base units will report deployed aircraft information on the main tab of the report. The deployment column will be labeled with "DEP" using the mission drop-down box. When a unit is supporting more than one deployment concurrently, each deployment will be reported under a separate column and clearly designated using standard operational or geographic

identifiers (e.g., DEP-CENTCOM, DEP-EUCOM). **(T-2) Note:** Exercises such as RED FLAG, GUNSMOKE, AIR WARRIOR, etc., are not reported as deployed operations.

3.7. Frequency of Reporting. Units will submit the 9302 report to HQ ACC/A4MO monthly.

3.7.1. Data and Driver sections are due No Later Than (NLT) 1600 EST on the 8th calendar day of the month following the reporting period (e.g., November data is due December 8th).

3.7.2. The Comments section is due NLT 1600 EST on the 10th calendar day.

3.7.3. Units will inform ACC/A4MO immediately upon realizing any situation that may cause a late submittal, prior to the established due date. Exercises, down-days, and standard inspections are not valid justifications for late submissions. **(T-2)**

3.8. Report Preparation. The report consists of three primary sections: Section 1 (Main Tab), Section 2 (Drivers Tab), and Section 3 (Comments Tab).

3.8.1. **Section 1 - Main Tab.** This tab contains the overall Wing, MDS, Block, and Mission data totals. A single unit's report may contain multiple MDS, Blocks, and/or Missions on this tab.

3.8.1.1. Enter the current reporting month-year in the Report Month cell; this will automatically update the date across all worksheets. Ensure deployed information is reported using the "DEP" mission identifier. **(T-2)**

3.8.1.2. **Deviation Reporting.** Refer to DAFI 21-101_ACCSUP, Attachment 8, for guidance on reporting specific deviations used in Fleet Standard Evaluation (FSE) rate calculations.

3.8.2. **Section 2 - Drivers Tab.** This tab contains subsystem drivers (three-digit Work Unit Code [WUC] / Logistics Control Number [LCN]) for the following rates: TNMCM, TNMCS, NMCM, NMCS, NMCB, Abort, Break, Fix, and Cannibalizations. All units will report on all drivers based on their assigned template. **(T-2) Note:** F-22 units lack a three-digit subsystem WUC/LCN and will use the two-digit system. F-35 units must include the letter prefix for consistency (e.g., the CNI System is “A48”).

3.8.2.1. Cannibalization drivers must include the full National Stock Number (NSN), part number (if obtained through a CLS contractor), full WUC/LCN, and accurate nomenclature. Stating "NSN not available" is strictly prohibited.

3.8.2.2. Reports will be returned for correction for the following discrepancies: incorrect data (e.g., possessed hours exceeding Total Aircraft Inventory [TAI]), lack of comments, unclear analysis, incorrect dates, repeated "copy/paste" verbiage from previous months, missing drivers, or cannibalization drivers lacking the correct subsystem/WUC/LCN. **(T-2)**

3.8.3. **Section 3 - Comments Tab.** The MXG/CC (or equivalent) will submit comprehensive narratives to the Director, HQ ACC/A4, detailing operational constraints, limiting factors, actionable data, and proposed solutions.

3.8.3.1. These comments must analyze the root causes for any missed standards and provide context for cumulative unit data.

3.8.3.2. Consolidate all remarks into a single narrative for each of the following three categories: Flying Hour Program (FHP) Execution, Overall Fleet Health, and MAJCOM Required Assistance. **(T-2)**

3.8.4. **FHP Execution (Comments).** This section provides a transparent overview of flying hour production, comparing planned objectives with contracted resources and actual execution.

3.8.4.1. **Annual Plan:** State the yearly flying hour program objective (e.g., 1250 sorties / 2500 hours). **(T-2)**

3.8.4.2. **Monthly Contract:** Detail the scheduled and contracted sorties/hours for the specific month. **(T-2)**

3.8.4.3. **Analysis (Plan vs. Contract):** Explain any discrepancies between the contracted amount and the annual plan. **(T-2)**

3.8.4.4. **Execution:** Report actual sorties and hours flown during the month. **(T-2)**

3.8.4.5. Analysis (Contract vs. Execution): Analyze the difference between contracted and executed hours, providing specific reasons for deviations. **(T-2)**

3.8.5. **Ready Aircrew Program (RAP) Status.**

3.8.5.1. **Met:** State if the unit met RAP requirements and briefly explain how. **(T-2)**

3.8.5.2. **Not Met:** State if the unit failed to meet RAP requirements, explain the operational impact on the remaining FHP, and provide a comprehensive "Get Well" plan. **(T-2)**

3.8.6. **Overall Fleet Health (Comments).** Provide specific examples regarding the ability to support missions and maintain readiness. Consider personnel, equipment, resources, and training. Highlight significant, multi-month trends. Relevant metrics include Mission Capable rate, TNMCM, TNMCS, Break/Fix rates, Repeat/Recur rates, and Cannibalization rates (IAW AFTTP 3-4.21, Volume 1, *Aircraft Maintenance*). **(T-2)**

3.8.7. **MAJCOM Required Assistance (Comments).** Detail areas of concern requiring support from the MAJCOM. Provide context valuable for long-term analysis, not just immediate problem-solving. If no assistance is required, state "NSTR" (Nothing Significant To Report). **(T-2)**

3.8.7.1. For inspections, Time Change Items, or Time Compliance Technical Orders/Directives (TCTO/TCTD), explicitly state the type of inspection and the TCTO/TCTD number. **(T-2)**

3.8.7.2. Comments should synthesize inputs from maintenance management and sortie production personnel (e.g., MXG Leadership, DO, AMU OICs, AMXS Commanders). **(T-2)**

3.8.7.3. Analysis should focus on in-depth constraints rather than isolated parts. If discussing systems, include the full WUC/LCN. **(T-2)**

3.8.7.4. AMU Data: While subordinate unit data is not required in the standard monthly report, units will maintain this data and provide it promptly when tasked by Weapons System Teams (WSTs) or staff agencies answering senior leadership inquiries. **(T-2)**

3.8.7.5. Manning/Training: Provide the complete Air Force Specialty Code (AFSC) alongside any explanation of manning or training shortfalls.

3.9. HOF. Maintenance Management Analysis (MMA) is the OPR for conducting the HOF assessment.

3.9.1. The purpose of the HOF is to track cyclical trends, review anomalies, and provide data for presentations, reports, and briefings.

3.9.2. Units will conduct the HOF assessment monthly. **(T-2)**

3.9.3. The HOF assessment will include, at a minimum: **(T-2)**

3.9.3.1. All MAJCOM-reportable leading and lagging indicators.

3.9.3.2. WUC/LCN analysis to identify driving systems.

3.9.3.3. Aircraft in Category 1, 2, and 3 Hangar Queen status, including corresponding "Get Well" plans.

3.9.3.4. Aircraft Engine and Pod status.

3.9.3.5. Results from special studies.

3.9.3.6. Seasonal trend analysis.

3.9.3.7. Abort counts and associated action items.

3.9.3.8. Results of implemented Courses of Action (COAs).

3.10. RIM Responsibilities.

3.10.1. **HQ ACC (A4MO) is responsible for:**

3.10.1.1. Providing oversight and guidance for RIM implementation.

3.10.1.2. Maintaining standardization of RIM calculations across subordinate units.

3.10.1.3. Coordinating with stakeholders to address systemic constraints.

3.10.1.4. Publishing the RIM North Star for each MDS/unit.

3.10.1.5. Conducting monthly MAJCOM RIM Reviews.

3.10.2. **NAF Commanders are responsible for:**

3.10.2.1. Conducting RIM reviews to assess wing-level fleet management constraints.

3.10.2.2. Maintaining awareness of subordinate wings' RIM data (e.g., AA rates, Maintenance/Supply/Depot rates).

3.10.2.3. Analyzing driving factors and identifying constraints impacting NAF fleets.

3.10.2.4. Collaborating with ACC/A4, Item Managers, and System Program Offices to address constraints based on RIM requirements.

3.10.2.5. Determining mandatory attendees for NAF RIM reviews (Recommended: NAF/A3, NAF/A4, Wing CCs, WSTs, PEOs, SPOs, SCOW, SCMW, Air Force Sustainment Center, and DLA). Stakeholders will be prepared to discuss mitigation efforts.

3.10.3. **Wing Commanders are responsible for:**

3.10.3.1. Implementing RIM guidance IAW this instruction and ensuring accurate data collection.

3.10.3.2. Incorporating RIM reviews into standard wing standup meetings. **(T-2)**

3.10.3.3. Briefing COMACC monthly on key field-level constraints, operational impacts, and specific areas requiring higher headquarters support. **(T-2)**

3.10.4. Data Submission and Timeline: All RIM data submissions are due to ACC/A4MO NLT 1600 EST on the 10th calendar day of each month. **(T-2)**

3.11. Analysis Tools.

3.11.1. **Purpose.** Utilizing standardized analysis tools ensures aircraft and systems are properly maintained and ready to support missions. Maintenance Managers will use these tools to provide commanders with accurate, actionable data to improve reliability, assess fleet health, and develop corrective actions.

3.11.2. **Analysis Products and Tools.** MMA will utilize various data analysis tools to support 9302 reporting and daily fleet assessments. At a minimum, Analysis sections will incorporate the following into standard processes:

3.11.2.1. **Control Charts:** Used to assess variability, detect trends, and ensure processes remain within control limits. Charts for variables (quantifiable data) and charts for attributes (categorical data) will be updated monthly to provide insights for continuous improvement.

3.11.2.2. **Seasonal and Historical Trend Analysis:** Analysts will use a minimum of four years of historical data to account for seasonal variations, removing outliers to provide a realistic assessment of recurring issues. This data will be updated monthly (following the finalization of the reported month's data) to predict future constraints and inform scheduling. **(T-2)**

3.11.2.3. **Root Cause Analysis (RCA):** Conducted when metrics fail to meet standards or abnormal trends are detected. Analysts will utilize tools such as Pareto Analysis, Cause-and-Effect (Fishbone) Diagrams, and Process Mapping to identify and address underlying problems rather than surface-level symptoms.

Chapter 4

DEPLOYMENT REQUIREMENTS

4.1. Applicability. This chapter establishes guidance and reporting requirements for all units executing deployment commitments in support of contingencies, exercises, crises, or wartime operations.

4.2. Reporting Requirements.

4.2.1. Deployed analysts will strictly comply with all reporting guidelines established by the gaining MAJCOM, Air Force Component Command, and Joint Task Force they are tasked to support.

4.2.2. Analysts will refer to and comply with applicable component command guidance (e.g., DAFI 21-101_USAFCENTSUP) for any additional, theater-specific reporting requirements. **(T-2)**

4.3. Deployed Analysis Preparation. Required Analyst deployment tools are explicitly outlined in the Mission Capability (MISCAP) Line Remarks and the Contingency Response Cell (CRC) guidance for each specific tasking.

4.3.1. Each unit will develop and maintain customized deployment tools for their MMA section to ensure robust support of their assets in theater.

4.3.2. Units will prepare for operations in Disconnected, Interrupted, and Low-Bandwidth (DIL) environments. Required tools include, but are not limited to, printed regulations, automated tracking spreadsheets, and comprehensive base Point of Contact (POC) lists. The specific list of requirements will be adapted based on the deployment scenario and physical location. **(T-2)**

4.3.3. Wing or command-level analysts currently operating at deployed locations are highly encouraged to contact their inbound replacements at the earliest opportunity. This turnover communication should clearly identify any specific training, system access, or job knowledge requirements necessary to successfully perform assigned duties at that specific location.

JON A. EBERLAN, Brigadier General, USAF
Director of Logistics, Engineering
and Force Protection

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFI 10-601, *Operational Capability Requirements Documentation and Validation*, 27 Apr 2021

AFI 16-402, *Aerospace Vehicle Programming, Assignment, Distribution, Accounting, and Termination,* 21 Aug 2026

AFI 21-101_ACCSUP, *Aircraft and Equipment Maintenance Management*, 23 Jun 2020

AFI 21-103_ACCSUP, *Equipment Inventory, Status, and Utilization Reporting*, 10 Dec 2020

AFI 33-322, *Records Management and Information Governance Program*, 23 Mar 2020

AFTTP 3-4.21V1, *Aircraft Maintenance,* 9 Feb 2022

DAFI 21-101, *Aircraft and Equipment Maintenance Management*, 20 Dec 2023

DAFI 21-103, *Equipment Inventory, Status, and Utilization Reporting*, 01 Nov 2022

DAFMAN 23-300, *Material Management Procedures,* 22 Jul 2025

DAFMAN 90-161, *Publishing Processes and Procedures,* 18 Oct 2023

DAFPD 10-9, *Lead Command/Lead Agent Designation and Responsibilities for United States Air Force Weapon Systems, Non-Weapon Systems, and Activities*, 25 May 2021

TO 00-20-2, *Maintenance Data Documentation*, 19 Dec 2023

Prescribed Forms

None

Adopted Forms

DAF Form 847, *Recommendation for Change of Product*

Abbreviations and Acronyms

AMU—Aircraft Maintenance Unit

AMXS—Aircraft Maintenance Squadron

CLS—Contractor Logistics Support

CND—Cannot Duplicate

COA—Course of Action

COMACC—Commander Air Combat Command

DDR—Data Documentation Record

DEP—Deployed

FCF—Functional Check Flight

FHP—Flying Hour Program

FMC—Fully Mission Capable

FSE—Flying Scheduling Effectiveness

HOF—Health of Fleet

JDD—Job Data Documentation

LCN—Logistics Control Number

LPI—Logistics Performance Indicator

MAJCOM—Major Command

MC—Mission Capable

MDS—Mission Design Series

MEFL—Minimum Essential Function List

MESL—Minimum Essential Subsystem List

MMA—Maintenance Management Analysis

MICAP—Mission Impaired Capability Awaiting Parts

MX—Maintenance

NLT—Not Later Than

NMC—Not Mission Capable

NMCM—Not Mission Capable Maintenance

NMCS—Not Mission Capable Supply

NMCB—Not Mission Capable Both

NSTR—Nothing Significant to Report

NSN—National Stock Number

OCF—Operational Check Flight

OPR—Office of Primary Responsibility

OPS—Operational

PA—Personnel Availability

PAI—Primary Aircraft Inventory

PDAI—Primary Development/Test Aircraft Inventory

PF—Phase Flow

PIC—Purpose Identifier Code

PM—Program Manager

PMAI—Primary Mission Aircraft Inventory

PMC—Partially Mission Capable

PMCS—Partially Mission Capable Supply

PMEL—Precision Measurement Equipment Laboratory

POAI—Primary Other Aircraft Inventory

POC—Point of Contact

PTAI—Primary Training Aircraft Inventory

RAP—Ready Aircrew Program

RCA—Root Cause Analysis

RCS—Report Control Symbol

RIM—Readiness Informed Metric

SIPRNET—Secret Internet Protocol Router Network

TAI—Total Aircraft Inventory

TCTD—Time Compliance Technical Directive

TCTO—Time Compliance Technical Order

TNMCM—Total Not Mission Capable Maintenance

TNMCS—Total Not Mission Capable Supply

TNG—Training

TES—Test

UTE—Utilization Rate

WST—Weapon System Teams

WUC—Work Unit Code

Terms

Adjusted Sorties Scheduled—The sum of total sorties scheduled (home base, off-station, deployed, and additions for end-of-fiscal-year flying hour program close-out) minus any utilization rate cancellations. The formula is located in AFI 21-101_ACCSUP, Attachment 8. This formula is utilized when calculating Flying Scheduling Effectiveness (FSE).

Air Abort—A sortie that could not continue its primary or alternate mission. An air abort for the same discrepancy during a Functional Check Flight (FCF) or Operational Check Flight (OCF) is not counted in the air abort rate. If the aircraft air aborts on a different component during the FCF/OCF, it is counted in the air abort rate. An In-flight Emergency (IFE) is a declaration of an emergency by the pilot during flight; an IFE does not constitute an air abort unless the primary or alternate mission could not be completed.

Air Abort Rate—A leading indicator of aircraft reliability. It represents the percentage of missions aborted in the air per 100 sorties flown. This rate only includes air aborts caused by Maintenance or Operations.

Aircraft Availability (AA)—The percentage of a fleet that is in a Mission Capable condition and not in a Depot-possessed status or Unit Possessed Non-Reportable (UPNR) possession. TO 00-20-2 outlines which Possession Identifier Codes (PICs) count toward AA. **Note:** This metric may be created at the Mission Design (MD)/MDS level or grouped by fleet (e.g., Aggregate, Bombers, Fighters).

Attrition (Aircraft)—Aircraft required to replace primary aircraft inventory losses. Do not confuse this term with computing scheduling attrition rates (e.g., weather-related cancellations) used when developing the flying schedule.

Average Possessed Aircraft (APA)—*See Sortie Usage Effectiveness (USE) Rate*.

Break—An aircraft that lands with an overall aircraft landing status of Code-3 based on the applicable aircraft MESL/MEFL. If an aircraft lands with an issue rendering it Not Mission Capable (NMC) per the MESL/MEFL, but the aircraft is still flyable (i.e., NMCMM), the Code-3 aircraft still counts toward the break rate. A break for the same discrepancy during an FCF/OCF does not count as a new break. If the aircraft breaks on a different component during the FCF/OCF, it is considered a new break.

Break Rate—A leading, flying-related metric representing the percentage of sorties that land with an overall aircraft landing status of Code-3. This metric primarily indicates aircraft system reliability and may reflect the quality of maintenance performed. It is an excellent predictor of parts demand. Trailing indicators include Mission Capable rate, TNMCS, Cannibalization Rate, and Repeat/Recur.

Cancellation (CX)—A scheduled sortie that could not fly its scheduled time and date prior to crew show. A cancellation deviation indicates the unit identified prior to execution that the scheduled flying plan would not be met.

Cannot Duplicate (CND)—The code utilized when reporting on-equipment maintenance when a discrepancy could not be recreated under test or troubleshooting conditions.

Cannot Duplicate (CND) Rate—The number of cannot duplicate discrepancies divided by total maintenance actions, multiplied by 100. See TO 00-20-2.

Cannibalization Rate—A leading indicator reflecting the number of cannibalization actions. It is calculated as the number of cannibalization removal actions divided by sorties flown, multiplied by 100. Cannibalizations are counted in the month the Action Taken code “T” (removal for installation) is processed. Per TO 00-20-2, depot aircraft cannibalization requires written Program Manager (PM) approval and counts toward this rate. A high Cannibalization Rate paired with a low TNMCS Rate may indicate improper use of the supply system and excessive induced maintenance man-hours.

Capability Code (CAP CODE)—The status of a system/subsystem at the end of a sortie, reflecting how the individual system performed and whether it was operated during the sortie.

Code-3—(Per DAFI 21-101) An aircraft or system with major discrepancies in mission-essential equipment that may require extensive repair or replacement prior to further mission assignment. The discrepancy may not affect safety-of-flight, and the aircraft may remain NMC flyable.

COMACC RIM Brief—A periodic briefing delivered to the ACC Commander detailing the status of the Readiness Informed Metric (RIM), highlighting key constraints, operational impacts, and required higher-headquarters support.

Contractor Logistics Support (CLS)—Non-government managed logistics processes (supply and/or maintenance) performed by civilian contractors intended to cover the total life cycle of a weapon system. Reference DAFI 63-101.

Contractor Logistics Support (CLS) and Government/CLS Partnerships—A collaborative arrangement utilizing contracted maintenance services to support Air Force operations.

Crew Ready—Indicates an aircraft is mission capable and ready for aircrew acceptance. The crew show phase begins once the aircrew is physically at the aircraft.

Crew Show—Indicates the aircrew has arrived at the aircraft and begun visual inspection. If an aircraft aborts after crew show and the "crew ready" spare is not accepted to generate the mission for any reason, the deviation is attributed to Operations.

Deferred (or Delayed) Discrepancy (DD)—Each deferred discrepancy is counted as either Awaiting Maintenance (AWM) or Awaiting Parts (AWP). Additional defer codes serve as sub-categories of AWM/AWP. DD defer codes are standardized by HAF/A4LM; local defer code listings are prohibited.

Deferred (or Delayed) Discrepancy Rate—A leading indicator representing the average deferred discrepancies across the unit's APA fleet (excluding awaiting depot input). It counts only AWM, AWM subcategories, and AWP.

Deviation—Any departure from the printed weekly flying schedule. This impacts the FSE rate. See status detail codes in AFI 21-103_ACCSUP.

Fix Rate—The percentage of sorties landing as Code-3 returned to a flyable status within a specified timeframe (clock hours), even if the aircraft remains NMC or NMCM flyable. Fix time stops when all Code-3 Pilot Reported Discrepancies (PRDs) are fixed. This metric measures repair speed and equipment maintainability.

Flying Scheduling Effectiveness (FSE) Rate—A leading indicator measuring how well a unit planned and executed the weekly flying schedule. It represents the percentage of sorties flown as scheduled. Deviations decreasing FSE include maintenance, supply, operations, weather, higher headquarters (HHQ), air traffic control, sympathy cancellations, early/late takeoffs, and additions.

Functional Check Flight (FCF)—A flight performed to ensure an aircraft is airworthy and capable of accomplishing its mission IAW applicable -6 TO requirements. An FCF with a System Cap Code 3 or 4 will not count toward the abort or break rate if it involves the specific system being tested.

Ground Abort—An aircraft that cannot take off due to maintenance, operations, supply, or weather.

Ground Abort Rate—The number of ground aborts (including spares) per sortie attempt, counting only those caused by Maintenance, Operations, and Supply. A ground abort for the same discrepancy during an FCF/OCF does not count toward this rate unless it involves a different component.

Hangar Queen (HQ)—A unit-possessed aircraft (including UPNR status) that has not flown for more than 30 consecutive days. See TO 00-20-2.

Hourly UTE Rate—The average hours flown per authorized aircraft/system per month.

Hourly USE Rate—The average hours flown per average possessed aircraft.

Joint Task Force—A command consisting of U.S. military personnel with a distinct regional or functional objective.

Line Remarks—Mission Capability (MISCAP) comments detailing specific tasking requirements.

Maintenance/Operations (MX/OPS) Deviation Rate—The number of MX/OPS deviations divided by adjusted sorties scheduled, multiplied by 100. This is a subset of FSE reflecting deviations within unit control.

Maintenance Scheduling Effectiveness (MSE)—A leading indicator measuring a unit's ability to plan and complete inspections and periodic maintenance on time. See AFI 21-101_ACCSUP, Attachment 9.

Mission Capable (MC) Rate—A lagging indicator measuring overall unit capability. It is the percentage of possessed hours for aircraft that are FMC or PMC.

Mission Impaired Capability Awaiting Parts (MICAP)—Expedited backorders for material required to satisfy immediate mission requirements. A "1A" MICAP is for an NMCS condition reportable in the 9302 report. A "JA" MICAP is for PMCS conditions.

Not Mission Capable Both (NMCB) Rate—The percentage of possessed time an aircraft/system is not mission capable due to both maintenance and supply.

Not Mission Capable Maintenance (NMCM) Rate—The percentage of possessed time an aircraft/system is not mission capable due to maintenance.

Not Mission Capable Supply (NMCS) Rate—The percentage of possessed time an aircraft/system is not mission capable due to supply.

Not Repairable This Station (NRTS)—Indicates an item cannot be repaired at the base level.

Operational Check Flight (OCF)—The first flight performed after extended downtime or extensive maintenance to validate a maintenance action.

Personnel Availability (PA)—Compares the number of authorized personnel (via the Unit Manning Document) to the number available for duty, excluding those reassigned, on leave, or TDY.

Phase Flow (PF) Average—A leading metric measuring the average phase time remaining on the fleet, typically targeted at half the inspection interval.

Pilot Reported Discrepancy (PRD)—Any maintenance event recorded against a sortie, including ground aborts, but excluding red ball maintenance events.

Possessed Aircraft—Aircraft under a wing commander's operational control and responsibility. Reference DAFI 21-103.

Primary Aircraft Inventory (PAI)—Aircraft assigned to meet primary aircraft authorization. See AFI 16-402.

Primary Mission Aircraft Inventory (PMAI)—Aircraft assigned for wartime missions (Possession Codes: CA, CC).

Primary Training Aircraft Inventory (PTAI)—Aircraft required for technical/specialized training (Possession Code: TF).

Primary Development/Test Aircraft Inventory (PDAI)—Aircraft assigned for testing and evaluation (Possession Code: CB).

Primary Other Aircraft Inventory (POAI)—Aircraft required for special missions not elsewhere classified.

Ready Aircrew Program (RAP)—A program ensuring aircrews maintain proficiency and currency, including flight hours, simulator training, and evaluations.

Recur Rate—A recurring discrepancy occurring on the second through fourth sortie (or attempted sortie) after corrective action. High rates may indicate rushed maintenance or poor troubleshooting.

Repeat Rate—A recurring discrepancy occurring on the very next sortie (or attempted sortie) after corrective action.

Repeat/Recur (R/R) Discrepancy Rate—A leading indicator tracking total repeat/recur malfunctions versus total aircrew discrepancies.

RIM (Readiness Informed Metric)—A systematic approach for calculating the quantity of aircraft resources required to achieve aircraft availability for training and operational requirements.

Scheduled Sortie—An aircraft scheduled for flight by tail number on the weekly schedule and confirmed on the daily schedule.

Sortie—Begins when an aircraft takes off and begins forward movement; ends when it returns to the surface and engines are stopped, it sits for over five minutes, or the crew changes.

Sortie Attempts—Includes sorties flown and ground aborts not spared.

Sortie Utilization Effectiveness (UTE) Rate—Average sorties flown per authorized aircraft per month.

Sortie Usage Effectiveness (USE) Rate—Average sorties flown per average possessed aircraft.

Spare Ground Abort—An event after crew show time preventing a crew-ready aircraft from flying, but the mission is saved by a spare. Counts in the Ground Abort Rate, but not against FSE.

Total Abort Rate—Total number of aborts (air, ground, spare ground) per sortie attempt.

Total Active Inventory (TAI)—Aircraft assigned to operating forces. See AFI 16-401.

Total Not Mission Capable Maintenance (TNMCM) Rate—Percentage of possessed time an aircraft is NMCM or NMCB.

Total Not Mission Capable Supply (TNMCS) Rate—Percentage of possessed time an aircraft is NMCS or NMCB.

Wing Stand-Up—A daily meeting led by the Wing Commander to assess aircraft status, maintenance progress, and parts availability.

Attachment 2

ALGORITHMS FOR LOGISTICS - MAINTENANCE PERFORMANCE INDICATORS

A2.1. Purpose of Maintenance Metrics. Maintenance metrics provide a quantifiable measurement of performance and capability. Maintenance managers will utilize these metrics to determine how well the unit is meeting mission requirements, to identify emerging support problems, and to project future trends.

A2.1.1. Metrics are a crucial form of information utilized by maintenance leaders to improve the performance of organizations, equipment, and personnel when measured against established goals and standards. Properly utilized, metrics serve as a roadmap to assess past performance, determine current trajectory, and validate the strategies required to achieve operational objectives.

A2.1.2. The overarching objective of Air Force maintenance is to maintain aircraft and equipment in a safe, serviceable, and ready condition to meet mission needs. Maintenance metrics support this objective. Headquarters Air Force (HAF), Major Commands (MAJCOMs), Wings, and Squadrons will establish and maintain metrics to evaluate and improve equipment condition, personnel skills, and long-term fleet health. Commanders at all echelons will utilize these metrics to drive performance improvements and ensure adherence to established guidelines**.**

A2.2. The Role of Analysis. Analysis is vital to improving organizational performance and is central to the management process. Commanders and maintenance managers will evaluate metrics objectively, relying on the Maintenance Management Analysis (MMA) section for unbiased, data-driven information.

A2.2.1. Effective maintenance managers will not strive to artificially “chase numbers.” Instead, they will utilize metrics to accurately allocate resources, assign personnel, and refine processes.

A2.2.2. Managers must thoroughly understand and communicate the connection between goals, standards, and metrics. The Air Force sets explicit goals and standards for organizations, personnel, and weapons systems to facilitate evaluation and continuous improvement. Senior leadership publishes these standards separately, and they must be clearly understood at all levels of command.

A2.2.3. While maintenance analysts track and present the data, leadership at all levels must actively support the analytical process and directly utilize metrics to drive performance improvements; this is inherently a leadership responsibility.

A2.3. Primary Maintenance Metrics. Metrics are categorized as either leading or lagging indicators. Leading indicators identify a problem in its initial stages, as they directly measure the factors impacting maintenance’s capability to provide resources to execute the mission. Lagging indicators follow leading indicators and confirm firmly established trends.

A2.3.1. Maintenance leaders must review sortie production and maintenance health constantly. They must be knowledgeable regarding the specific maintenance indicators that highlight negative trends before those trends degrade operational capability.

A2.3.2. Technical Order (TO) 00-20-2 serves as the official, authoritative source for all maintenance metric definitions and formulas.

A2.3.3. **Maintenance Metrics Algorithms:** Personnel will utilize TO 00-20-2 as the sole source for calculating maintenance metrics. For specific code guidance, refer to DAFI 21-103, Attachment 2 (for maintenance status codes) and TO 00-20-2, Table 11-1 (for flying codes).

A2.3.4. **Average Possessed Number of Aircraft/Systems (APA):**

$$\frac{TOTAL\ POSSESSED\ HOURS\ (BEGINNING\ OF\ MONTH\ TO\ PREVIOUS\ DAY)}{(\#\ OF\ DAYS\ (BEGINNING\ OF\ MONTH\ TO\ PREVIOUS\ DAY)) \times 24}$$

A2.3.5. **Pre-Maintenance Time.**

$$\frac{TOTAL\ NUMBER\ OF\ DAYS\ IN\ PRE\ MAINTENANCE}{TOTAL\ NUMBER\ OF\ ITEMS\ IN\ TURNED\ IN}$$

A2.3.5.1. **Post-Maintenance Time.**

$$\frac{TOTAL\ NUMBER\ OF\ DAYS\ IN\ POST\ MAINTENANCE}{TOTAL\ NUMBER\ OF\ ITEMS\ IN\ TURNED\ IN}$$

A2.3.5.2. **Repair Time.**

$$\frac{TOTAL\ NUMBER\ OF\ DAYS\ IN\ REPAIR}{TOTAL\ NUMBER\ OF\ ITEMS\ IN\ TURNED\ IN}$$

A2.3.6. **Flying Scheduling Effectiveness Deviation Categories.** These rates reflect the sum of all Calculated Deviations in the various cause code categories. All deviations are defined per AFI 21-101_ACCSUP.

A2.3.6.1. **Maintenance (MT_ & GA_) Deviation Rate.**

$$\frac{TOTAL\ MAINTENANCE\ DEVIATIONS}{ADJUSTED\ SORTIES\ SCHEDULED} \times 100$$

A2.3.6.2. **Operations (OP_) Deviation Rate.**

$$\frac{TOTAL\ OPERATIONS\ DEVIATIONS}{ADJUSTED\ SORTIES\ SCHEDULED} \times 100$$

A2.3.6.3. **Supply (SU_) Deviations Rate.**

$$\frac{TOTAL\ SUPPLY\ DEVIATIONS}{ADJUSTED\ SORTIES\ SCHEDULED} \times 100$$

A2.3.6.4. **Higher Headquarters (HQ_) Deviation Rate.**

$$\frac{TOTAL\ HHQ\ DEVIATIONS}{ADJUSTED\ SORTIES\ SCHEDULED} \times 100$$

A2.3.6.5. **Weather (WX_) Deviation Rate.**

$$\frac{TOTAL\ WEATHER\ DEVIATIONS}{ADJUSTED\ SORTIES\ SCHEDULED} \times 100$$

A2.3.6.6. **Sympathy Deviation (SY_) Rate.**

$$\frac{TOTAL\ SYMPATHY\ DEVIATIONS}{ADJUSTED\ SORTIES\ SCHEDULED} \times 100$$

A2.3.6.7. **Air Traffic Control (ATC) Deviation Rate.**

$$\frac{TOTAL\ ATC\ DEVIATIONS}{ADJUSTED\ SORTIES\ SCHEDULED} \times 100$$

A2.3.6.8. **Other (OT_) Deviation Rate.**

$$\frac{TOTAL\ OTHER\ DEVIATIONS}{ADJUSTED\ SORTIES\ SCHEDULED} \times 100$$

A2.3.6.9. **UTE Deviation Rate.**

$$\frac{TOTAL\ UTE\ DEVIATIONS}{ADJUSTED\ SORTIES\ SCHEDULED} \times 100$$

A2.3.7. **Maintenance Man-hours per Flying Hour.**

$$\frac{TOTAL\ DIRECT\ MAN\ HOURS\ DOCUMENTED\ AGAINST\ AIRCRAFT\ AND\ ENGINES}{TOTAL\ HOURS\ FLOWN}$$

Note: Excludes PMEL, AGE, SE, and Transient Maintenance Hours.

A2.3.8. **Monthly JDD Error Rate after Corrections.**

$$\frac{(TOTAL\ DDRS\ IN\ ERROR - TOTAL\ DDRS\ CORRECTED)}{TOTAL\ DDRS\ IN\ ERROR} \times 100$$

A2.3.9. **Monthly JDD Error Rate.**

$$\frac{TOTAL\ DDRS\ IN\ ERROR}{TOTAL\ DDRS\ CHECKED} \times 100$$

A2.3.10. **Phase Flow (PF) Rate.**

$$\frac{TOTAL\ HOURS\ OF\ ALL\ POSSESSED\ AIRCRAFT\ UNTIL\ NEXT\ PHASE}{TOTAL\ AIRCRAFT\ INVENTORY}$$

Attachment 3

READINESS INFORMED METRICS

A3.1. Purpose. This attachment establishes the Readiness Informed Metrics (RIM) for calculating aircraft requirements for all ACC units. RIM ensures adequate aircraft availability to meet operational, training, and maintenance demands.

A3.2. RIM Formulas. The MAJCOM will conduct an annual review of the RIM formulas to assess their continued applicability and evaluate the validity of incorporated factors. **(T-2)**

A3.2.1. **RIM Aircraft Availability Calculation.** The Aircraft Availability is calculated under the assumption that all aircraft utilized are mission capable. **(T-2)**

A3.2.2. **RIM Aircraft Availability Standard Factor Considerations.** Any of the following defined factors, alongside other relevant operational and mission variables, may be considered for the Aircraft Availability formula:

A3.2.2.1. **O&M Days:** The allocated number of Operations and Maintenance days within the fiscal year. O&M days are Monday through Friday, not including federal holidays or command directed family days. **Note:** ACC will release a listing of federal holidays and command directed family days prior to the start of a new calendar year.

A3.2.2.2. **Turn Rate:** A factor representing the average sortie generation capability of the fleet.

A3.2.2.3. **Attrition:** A factor that represents the historical percentage of scheduled sorties lost due to causes outside unit control. It is not intended to substitute for unit capability shortfalls but is added to mitigate scheduling turbulence and ensure the unit's mission goals are met. The attrition computation is derived from unit historical data for previous, comparable flying months. For instance, attrition for January 12 would be based on historical data from January 11, January 10, January 09, January 08, and so on, using a minimum of four years of data to account for seasonal variations.

A3.2.2.4. **Pilot Proficiency Requirement:** The number of aircraft required to support the sorties required to maintain proficiency for combat-coded pilots.

A3.2.2.5. **Spare Requirements:** The number of aircraft required daily to maintain adequate spare aircraft inventory for the entire fleet throughout the fiscal year.

A3.2.2.6. **Ground Requirements:** The number of aircraft required daily to fulfill ground training requirements for the entire fleet throughout the fiscal year.

A3.2.2.7. **Fleet Management Requirement:** The number of aircraft excluded from daily flying schedules to facilitate fleet health management.

A3.2.2.8. **Alert Requirement:** The number of aircraft required daily to meet alert posture requirements for the entire fleet throughout the fiscal year. Alert status mandates immediate readiness of aircraft, spacecraft, and crews for takeoff or launch, typically within minutes, in response to unforeseen, time-sensitive situations.

A3.2.2.9. **Loaner:** The number of aircraft loaned out to other units.

A3.2.2.10. **Demo:** The number of aircraft required to support a Demo Team or other unit-specific requirements that have not been addressed in another category.

A3.2.2.11. **Contingency:** The number of aircraft required to support contingency requirements each day for the fiscal year.

A3.3. Standards Development for Readiness Indicators. To accurately support and validate the Readiness Informed Metrics (RIM), HQ ACC/A4MO, in coordination with Weapons System Teams (WSTs), will develop and publish performance standards for all leading and lagging maintenance indicators IAW DAFI 21-103. These standards establish the minimum acceptable performance thresholds required to sustain fleet viability and fulfill operational commitments.

A3.3.1. **Leading Indicator Standards.** Leading indicators (e.g., Break Rate, Fix Rate, Cannibalization Rate, Repeat/Recur Rate, and Flying Scheduling Effectiveness) serve as predictive measures of fleet health and maintenance execution. ACC/A4MO will engineer standards for these metrics specifically to identify early constraints in sortie generation, equipment reliability, and parts demand. Commanders will utilize these leading standards to preemptively adjust resources and maintenance practices before overall Aircraft Availability degrades.

A3.3.2. **Lagging Indicator Standards.** Lagging indicators (e.g., Mission Capable Rate, Non Mission Capable Maintenance, Non Mission Capable Supply, and Non Mission Capable Both) measure the historical culmination of maintenance and supply effectiveness. Standards for lagging indicators will directly align with the overarching RIM operational requirement, ensuring the fleet consistently produces the required AA to meet combat, test, and training taskings.

A3.3.3. **Annual Standard Review.** HQ ACC/A4MO will conduct an annual comprehensive review of all leading and lagging indicator standards. This review will analyze long-term historical performance, aging fleet dynamics, and shifting operational requirements. ACC/A4MO will publish the standards prior to the start of the new fiscal year to ensure total alignment with the updated AA standard.

A3.3.4. **Actionable Thresholds and Triggers.** Standards will serve as operational triggers rather than mere reporting benchmarks. When unit performance fails to meet established standards for leading or lagging indicators, the Maintenance Management Analysis (MMA) section will immediately initiate Root Cause Analysis (RCA) IAW **paragraph 3.11.2.3**.

A3.3.5. Units will utilize these standard deviations to drive actionable "Get Well" plans, redirect organic resources, and explicitly justify requests for MAJCOM or Lead Command assistance within their monthly 9302 report narratives. **(T-2)**